Heroes or Villains: The Role of Media in Shaping Public Perception of Ethical Hackers

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Abstract - This study examines the ways in which media representations of hacking affect public awareness and trust in ethical hackers (penetration testers) in comparison to malicious hackers. While the cybersecurity and technology fields set clear distinctions between "white hat" and "black hat" hacking, popular media typically blurs the line, presenting hackers primarily as malicious or criminal individuals. To address this, I conducted a survey to gather insights from everyday citizens on their personal perceptions of hackers and the media portrayals that shaped those views, complemented by a literature review and content analysis of mainstream and social media. Preliminary results show that media perceptions link ethical hackers with malicious hackers and contribute to a general lack of awareness regarding them. Despite penetration testers' vital role in protecting systems and data, this negative representation diminishes public trust in them. Raising awareness of ethical hackers is crucial as it can increase public trust in cybersecurity experts, drive individuals to learn more, and ultimately reduce the likelihood of malicious hacking. This study emphasizes the importance of fair representation and public education in cybersecurity by stressing the role of the media.

I. Introduction

In today's era of technology everything is connecting to The rapid growth of cybersecurity threats has made hacking a frequent subject of public discussion, yet perceptions of hackers remain heavily shaped by media portrayals rather than by technical realities. Popular films, television shows, and video games often depict hackers as malicious and manipulative criminals who thrive on deception and chaos. For example, TV shows such as *Power Book II: Ghost* introduce Effie Morales as a character who leverages hacking for manipulation and survival, reinforcing the association between hacking and criminal activity. Movies such as *Blackhat* and *Mr. Robot* portray the same. In gaming, *Grand Theft Auto V* features Lester Crest, a disabled hacker who orchestrates high-profile heists, while titles such as *Watch Dogs* and *Cyberpunk 2077* glamorize the hacker as an outlaw vigilante capable of bending entire cities to their will.

These portrayals overshadow the important role of ethical hackers, also known as penetration testers, who use their technical skills to identify and fix vulnerabilities in computer systems before malicious hackers can exploit them. Ethical hackers operate with permission and follow professional codes of conduct, helping organizations protect sensitive data and infrastructure. Unlike malicious hackers, or "black-hat" hackers, who seek to steal, disrupt, or damage systems, ethical hackers act to strengthen cybersecurity defenses and maintain trust in digital systems.

As Staggs, McMichael, and Kwan (2020) argue, repeated exposure to media representations shapes audience perceptions, often blurring the lines between legitimate cybersecurity work and illicit activity. Consequently, the public frequently conflates ethical hackers with malicious ones, creating a climate of mistrust toward even those professionals working to protect organizations. Research in the field emphasizes that ethical hackers play a pivotal role in safeguarding organizations from vulnerabilities, but this nuance rarely translates into mainstream portrayals. Hatfield (2019) highlights the ethical dimension of penetration testing, showing that "human hacking" techniques such as social engineering can be used responsibly when guided by professional codes of conduct. Similarly, Thomas, Burmeister, and Low (2019) stress that the trustworthiness of penetration testers becomes even more important in the era of strict breach disclosure laws, where organizations depend on their integrity to uncover weaknesses before criminals can exploit them. Despite this, characters like Effie, Lester, or the vigilante hackers in *Watch Dogs* are more widely

recognized than real-life ethical hackers, leaving the public with skewed assumptions about hacking as inherently criminal or morally ambiguous.

Therefore, this study investigates how media representations of hacking influence awareness and trust in ethical hackers compared to malicious hackers. By combining a literature review, a survey of public opinion, and a content analysis of mainstream and digital media, the research seeks to understand how these portrayals impact trust in cybersecurity professionals. The findings aim to highlight the consequences of inaccurate or sensationalized representations and underscore the importance of educating the public about the value of ethical hacking. Improving these perceptions is not only essential for building trust in cybersecurity professionals but also for fostering stronger collaboration between the public, organizations, and the individuals responsible for defending against cyber threats.

II. Methodology

This study utilizes a mixed-method approach to examine how media representations of hacking influence public perceptions of ethical hackers. It combines a literature review, a survey of everyday citizens, and a content analysis of mainstream and social media portrayals. The survey gathers insights on participants' familiarity with ethical hacking, exposure to media depictions of hackers, and trust in cybersecurity professionals. This approach allows for a comprehensive understanding of both media trends and public perception.

A. Target Audience

The primary target audience for this study is everyday citizens aged 18 and older who interact with media regularly, including TV, movies, video games, and social media. The survey aims to reach all individuals regardless of their background, occupation, or level of technical knowledge, to capture a broad understanding of public perceptions of hackers.

B. Research Approach

This study uses a mixed-methods approach, combining:

Literature Review: Analyze existing research on media influence, public perception, and the role of ethical hackers.

Survey Research: Develop and distribute a questionnaire using Microsoft forms to gather participants' personal perceptions of hackers and the media they consume.

Survey distribution: Surveys will be posted online via social media platforms (e.g., Facebook, Instagram, WhatsApp, Reddit), university forums, and other networks to reach a diverse audience.

Survey focus: Questions will explore participants' familiarity with ethical hackers, exposure to hacker portrayals in media, and trust in cybersecurity professionals.

Content Analysis: Examine mainstream and social media examples of hacker portrayals to identify recurring themes and stereotypes.

This approach allows the study to compare public perceptions with media trends and evaluate how media exposure shapes trust in ethical hackers.

C. Literature Review

[1]. "*Ethical hacking and its role in cybersecurity.*" Asif et al. (2024).

This article was chosen because it provides a detailed overview of ethical hacking practices and their significance in modern cybersecurity. It explains the methods penetration testers use to detect vulnerabilities, which helps clarify the technical aspects for readers unfamiliar with the field. The study highlights the real-world impact of ethical hackers in protecting organizations from cyberattacks. It also distinguishes ethical hacking from malicious hacking, reinforcing the foundational definitions for this research. Overall, it gives the study a solid technical framework to compare public perceptions against the reality of ethical hacking work.

[2]. "*Wishing to be like the character on screen: Media exposure and perception of hacking behavior*." Staggs, McMichael, & Kwan (2020).

This research was selected because it directly investigates how media shapes public perceptions of hacking behavior. It shows that repeated exposure to fictional hacker characters can influence viewers' understanding of real-world cybersecurity practices. The article supports the argument that media portrayals may blur the line between ethical and malicious hacking. It provides evidence that public attitudes are not always based on factual knowledge but are often shaped by entertainment media. This helps justify the focus of this study on media influence in shaping perceptions of ethical hackers.

[3]. "*Virtuous human hacking: The ethics of social engineering in penetration-testing.*" Hatfield (2019).

This article was included to highlight the ethical dimension of penetration testing. It explains how "human hacking" techniques, such as social engineering, can be conducted responsibly under professional codes of conduct. The research contrasts ethical hacking with malicious hacking, providing clarity for this study's definitions. It also emphasizes the role of integrity in cybersecurity, which ties directly to public trust. By

including this source, the study can better explain how ethical hackers operate, countering misconceptions created by media portrayals.

[4]. "*The importance of ethical conduct by penetration testers in the age of breach disclosure laws.*" Thomas, Burmeister, & Low (2019).

This article was chosen because it focuses on the critical role of trustworthiness among penetration testers. It demonstrates how organizations rely on ethical hackers to uncover vulnerabilities before malicious actors can exploit them. The study reinforces why public understanding and trust in ethical hackers are important. It provides examples of ethical standards and legal frameworks guiding professional conduct. Including this source helps link the research question to real-world consequences of public misconceptions about hackers.

[5]. "*Emerging ethical issues in cybersecurity and AI: A global perspective.*" Frontiers Editorial Team (2025).

This source was selected for its broader discussion of societal and ethical issues in cybersecurity and AI. It highlights how public perception can affect the adoption of security measures and trust in professionals. The article situates ethical hacking within larger debates about technology, ethics, and regulation. It supports the study's focus on media influence as a factor in shaping public awareness and trust. By referencing this work, the study connects individual perceptions to global trends and ethical considerations in cybersecurity.

III. References

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